The Physical Properties of boron allotrope

o-B14 and its mono-/bi-layer

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Overview

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Structures quick look

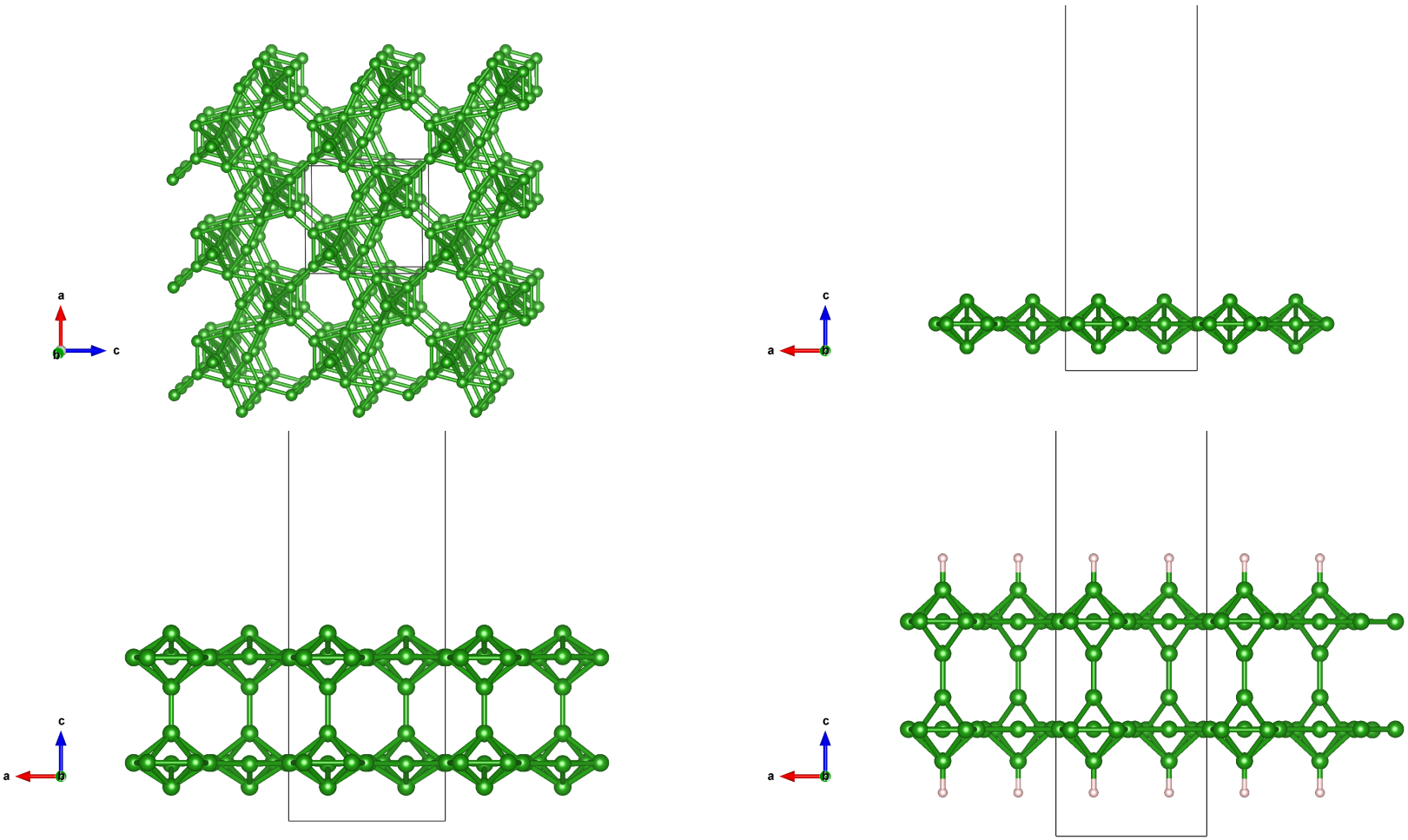

Figure 1: The quick look of structures.

Phonon dispersion

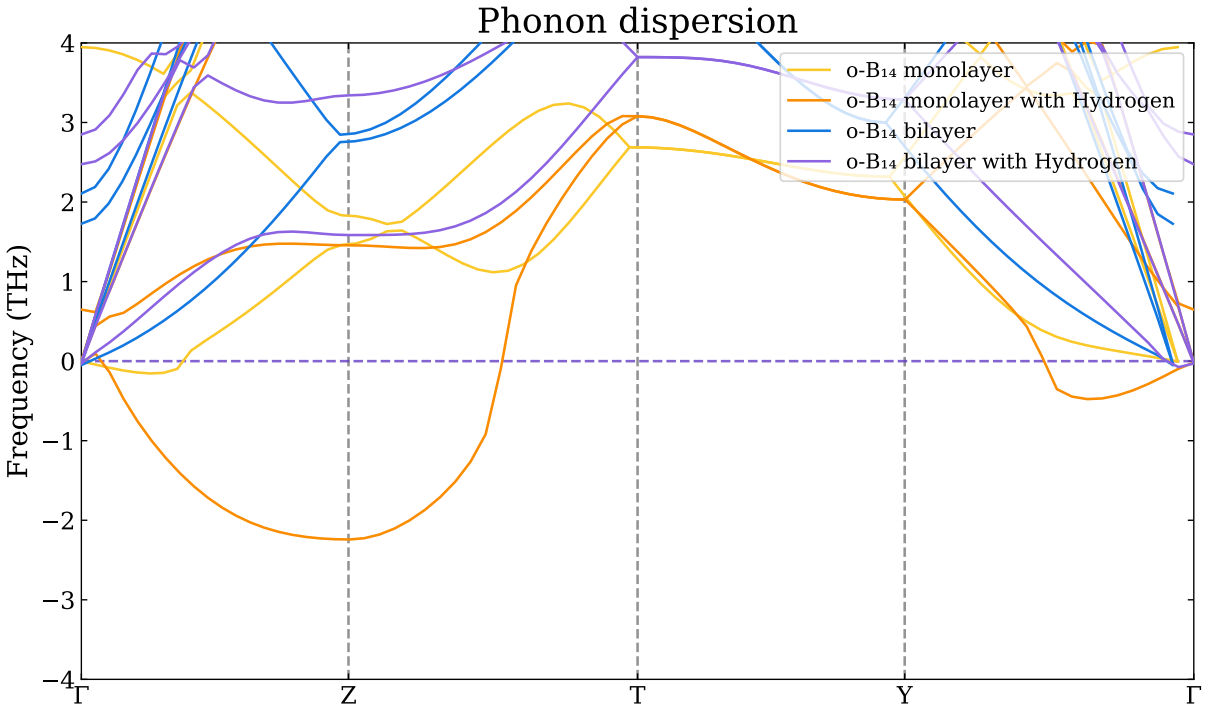

Figure 2: The Phonon dispersion calculations for monolayers and bilayers.

Bandstructure

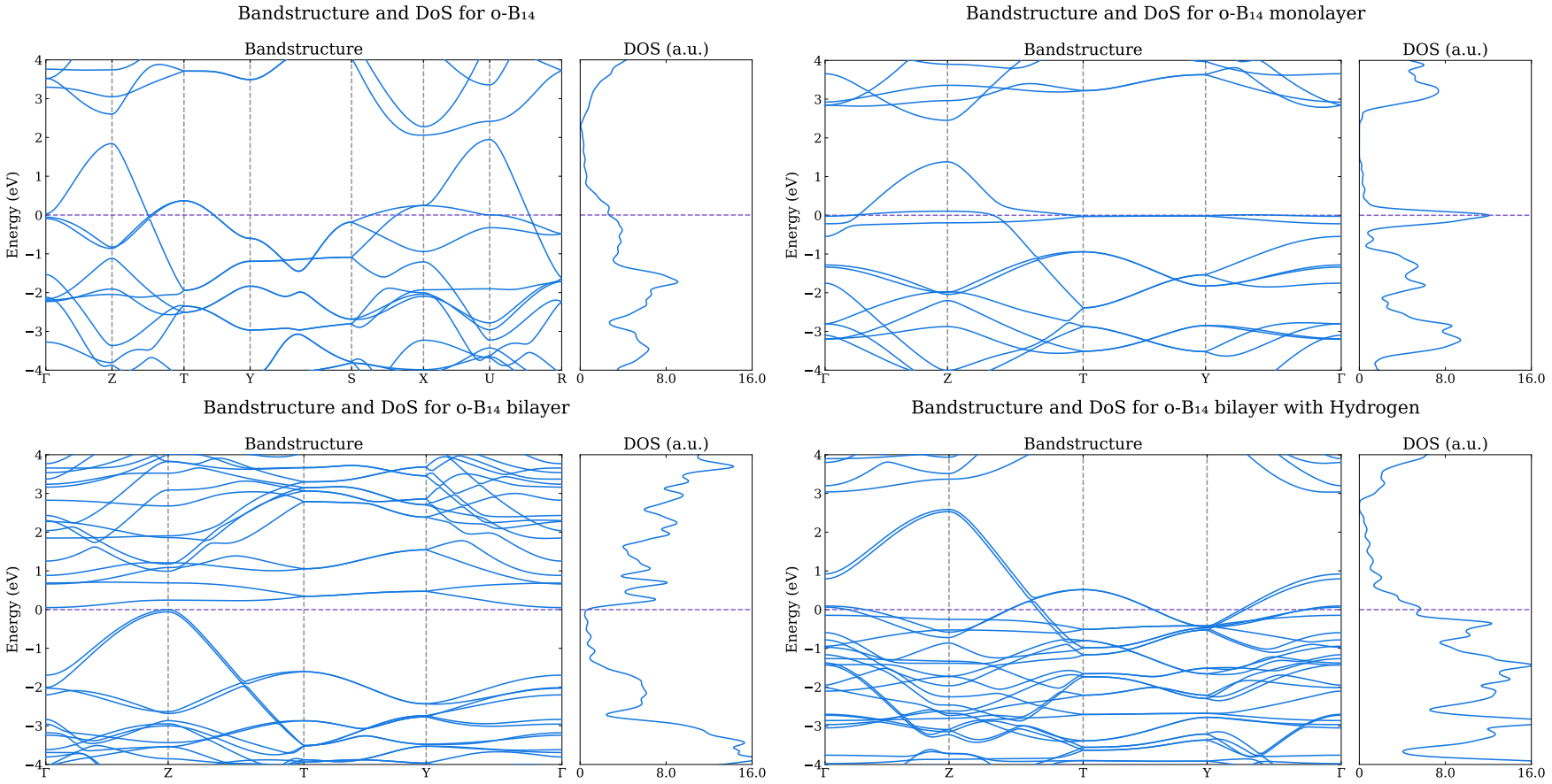

Figure 3: The band structure of the monolayers and bilayers

Density of state

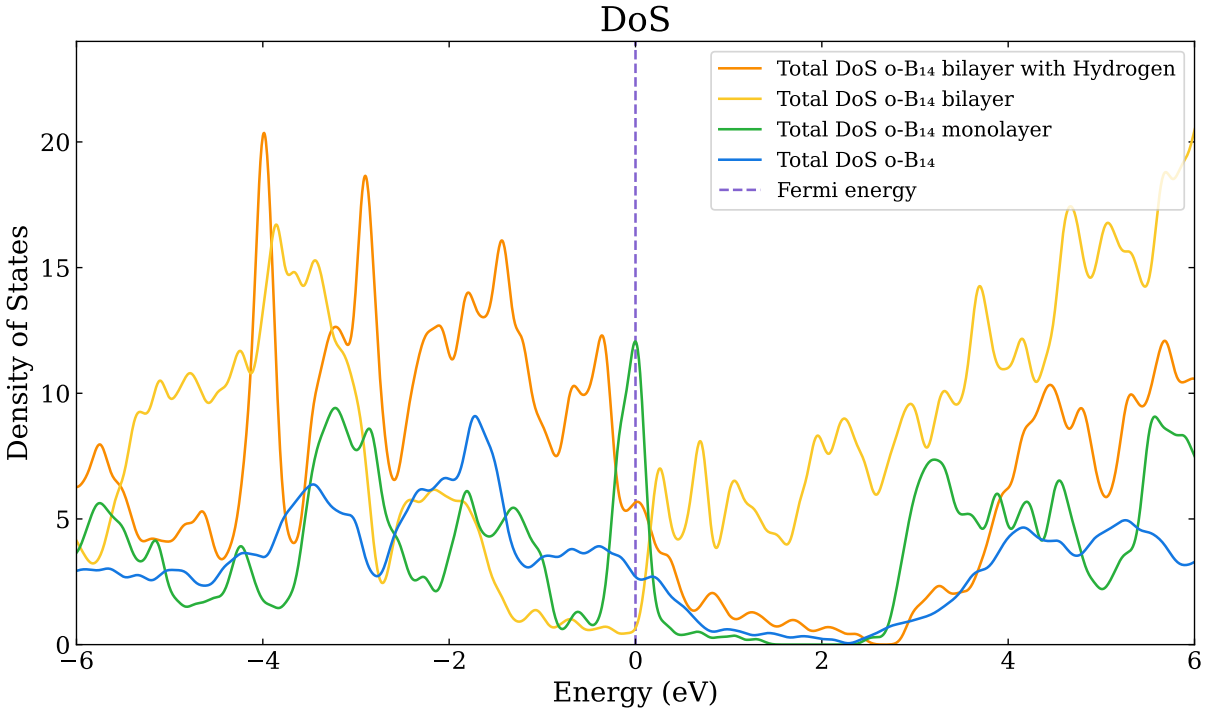

Figure 4: The electronic density of state.

Dielectric function

We investigated the dielectric function $\varepsilon(\omega)$ because understanding the dielectric function is fundamental to studying the optical performance of the system. It is an important physical quantity that describes the response of a system under an external electric field. It represents the degree of polarization of internal charges in the material under the influence of an electric field (or photon) and is a function of frequency ω. It can be obtained through electronic structure calculations.

$$\varepsilon(\omega) = \varepsilon_1(\omega) + i\varepsilon_2(\omega) \tag{1}$$

Where:
$\varepsilon_1(\omega)$ is the real part of the dielectric function, representing the system's energy storage capability.
$\varepsilon_2(\omega)$ is the imaginary part of the dielectric function, representing the system's loss or energy absorption capability.

The real part of the dielectric function $\varepsilon_1(\omega)$ can be expressed by the following formula:

$$\varepsilon_1(\omega) = \mathrm{Re}\ \varepsilon_{\alpha\beta}(\omega) = \delta_{\alpha\beta} + \frac{2}{\pi} P \int_0^\infty \mathrm{d}\omega' \frac{\omega' \varepsilon_2(\omega')}{\omega'^2 - \omega^2} \tag{2}$$

Here, P denotes the Cauchy principal value, indicating that the integral excludes the singularity when $\omega' = \omega$.
And $\frac{\omega' \varepsilon_2(\omega')}{\omega'^2 - \omega^2}$ the integrand, $\varepsilon_2(\omega)$ is the imaginary part of the dielectric function. This part of the expression represents the contribution of the imaginary part to the real part of the dielectric function.
The integral part of the formula shows how the imaginary part of the dielectric function $\varepsilon_2(\omega)$ influences the real part through the Kramers-Kronig relations.

The imaginary part of the dielectric function $\varepsilon_2(\omega)$ can be expressed by the following formula:

$$\begin{aligned}\varepsilon_2(\omega) =& \mathrm{Im}\ \varepsilon_{\alpha\beta}(\omega) \\ =& \frac{\hbar^2 e^2}{\pi m^2 \omega^2} \sum_n \\ & \int \mathrm{dk}\, \langle \psi_{\mathrm{k}}^{\mathrm{C}_n} | \mathrm{P}^\alpha | \psi_{\mathrm{k}}^{\mathrm{V}_n} \rangle \langle \psi_{\mathrm{k}}^{\mathrm{V}_n} | \mathrm{P}^\beta | \psi_{\mathrm{k}}^{\mathrm{C}_n} \rangle\, \delta(E_{\mathrm{k}}^{\mathrm{C}_n} - E_{\mathrm{k}}^{\mathrm{V}_n} - \omega) \quad (3)\end{aligned}$$

Where:
$\langle \psi_{\mathrm{k}}^{\mathrm{C}_n} | \mathrm{P}^\alpha | \psi_{\mathrm{k}}^{\mathrm{V}_n} \rangle$ and $\langle \psi_{\mathrm{k}}^{\mathrm{V}_n} | \mathrm{P}^\beta | \psi_{\mathrm{k}}^{\mathrm{C}_n} \rangle$ are the matrix elements of the momentum operator P in the α and β directions; $\psi_{\mathrm{k}}^{\mathrm{C}_n}$ and $\psi_{\mathrm{k}}^{\mathrm{V}_n}$ are the wave functions of the conduction band and valence band, respectively.

The calculation result:

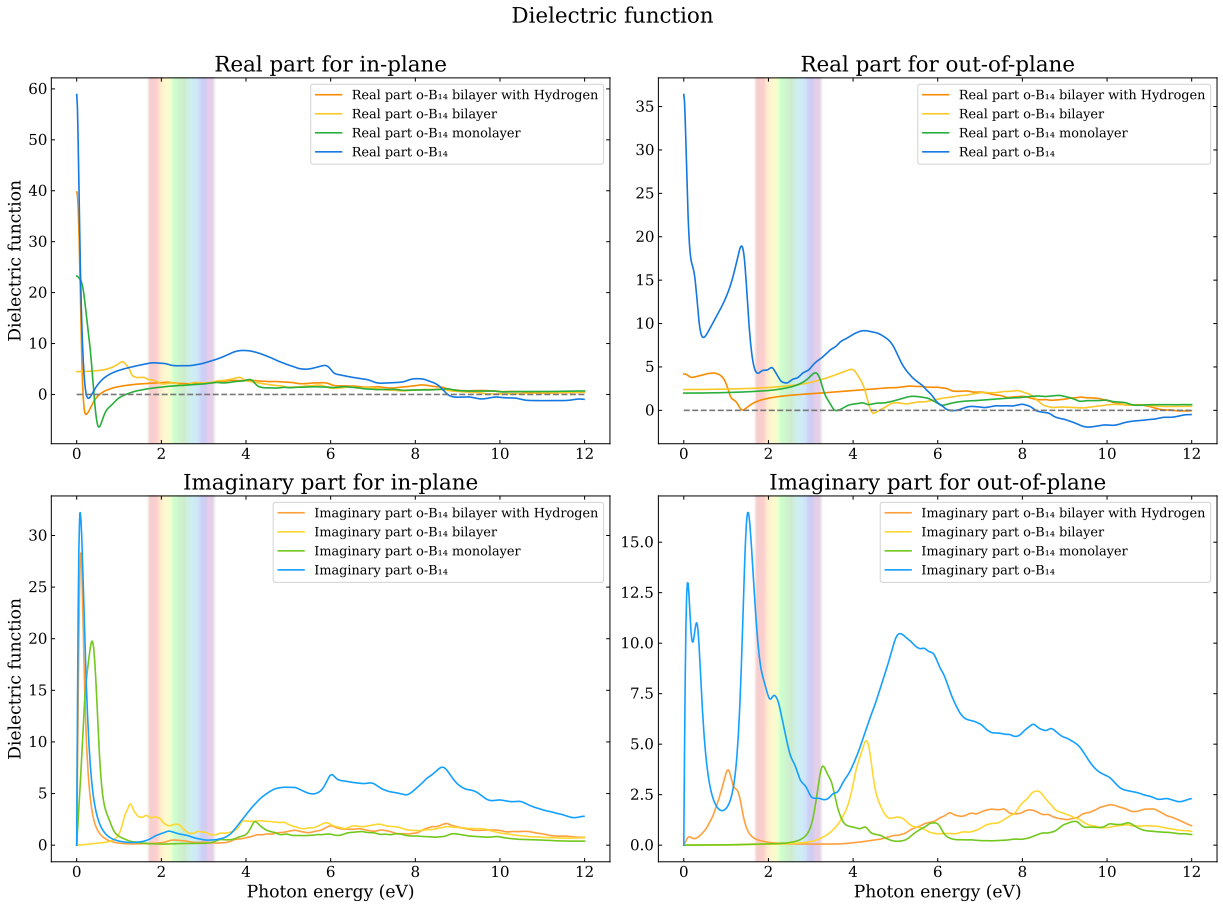

Figure 5: The in-plane and out-of-plane dielectric function.

Kramers-Kronig relation

The Kramers-Kronig relations describe the relationship between the real and imaginary parts of the frequency response of a causal system. Specifically, these relations indicate that if the imaginary part $\varepsilon_2(\omega)$ of the dielectric function is known, the real part $\varepsilon_1(\omega)$ can be obtained through an integral, and vice versa.
If $\varepsilon_2(\omega)$ is known, $\varepsilon_1(\omega)$ can be calculated using the following formula:

$$\varepsilon_1(\omega) = 1 + \frac{2}{\pi}\mathrm{P}\int_0^\infty \frac{\omega'\varepsilon_2(\omega')}{\omega'^2 - \omega^2}d\omega' \tag{4}$$

Conversely, if $\varepsilon_1(\omega)$ is known, $\varepsilon_2(\omega)$ can be calculated using the following formula:

$$\varepsilon_2(\omega) = -\frac{2\omega}{\pi}\mathrm{P}\int_0^\infty \frac{\varepsilon_1(\omega')}{\omega'^2 - \omega^2}d\omega' \tag{5}$$

Optical property: adsorption coefficient

From the complex form of the dielectric function, we know $\varepsilon(\omega) = \varepsilon_1(\omega) + i\varepsilon_2(\omega)$. The complex dielectric function can be used to calculate the complex refractive index is $N(\omega) = n(\omega) + ik(\omega)$, where $n(\omega)$ is the refractive index, and $k(\omega)$ is the extinction coefficient. Then, we can get $\alpha(\omega) = \frac{2\omega k(\omega)}{c}$, and:

$$\alpha(\omega) = \frac{\sqrt{2}\omega}{c}\left[\left(\varepsilon_1^2(\omega) + \varepsilon_2^2(\omega)\right)^{\frac{1}{2}} - \varepsilon_1(\omega)\right]^{\frac{1}{2}} \tag{6}$$

Optical property: adsorption coefficient

The calculation result:

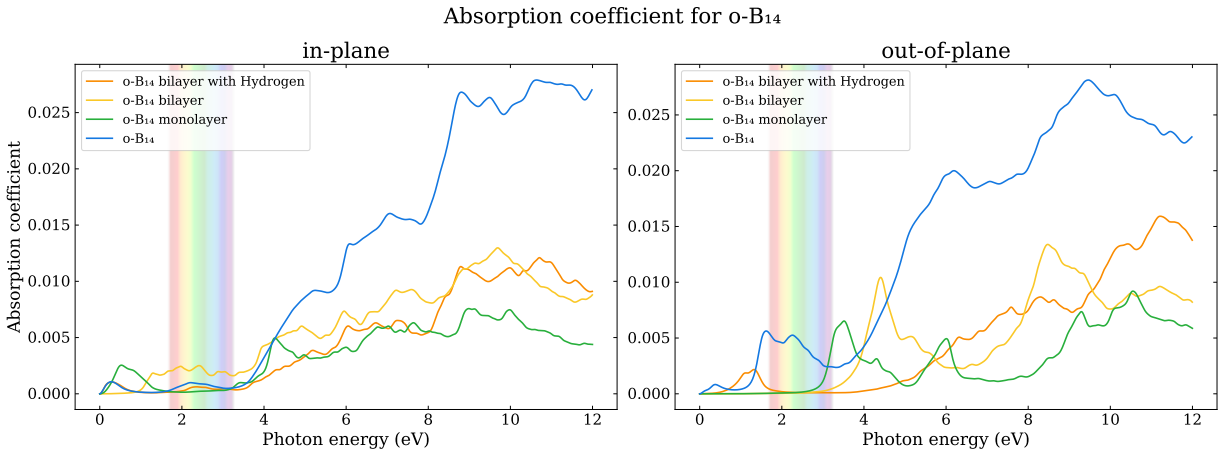

Figure 6: The in-plane and out-of-plane adsorption coefficient.

Optical property: refractive index and extinction coefficient

The refractive index and extinction coefficient describe the speed and attenuation of light in the system. From the complex dielectric function, the complex refractive index $N(\omega)$ can be define:

$$N(\omega) = n(\omega) + ik(\omega) \tag{7}$$

and it has a relation with the dielectric function $N(\omega)^2 = \varepsilon(\omega)$. Solve them, we can get the refractive index and extinction coefficient respectively:

$$n(\omega) = \left[\frac{\sqrt{\varepsilon_1(\omega)^2 + \varepsilon_2(\omega)^2} + \varepsilon_1(\omega)}{2} \right]^{\frac{1}{2}} \tag{8}$$

$$k(\omega) = \left[\frac{\sqrt{\varepsilon_1(\omega)^2 + \varepsilon_2(\omega)^2} - \varepsilon_1(\omega)}{2} \right]^{\frac{1}{2}} \tag{9}$$

The calculation result of refractive:

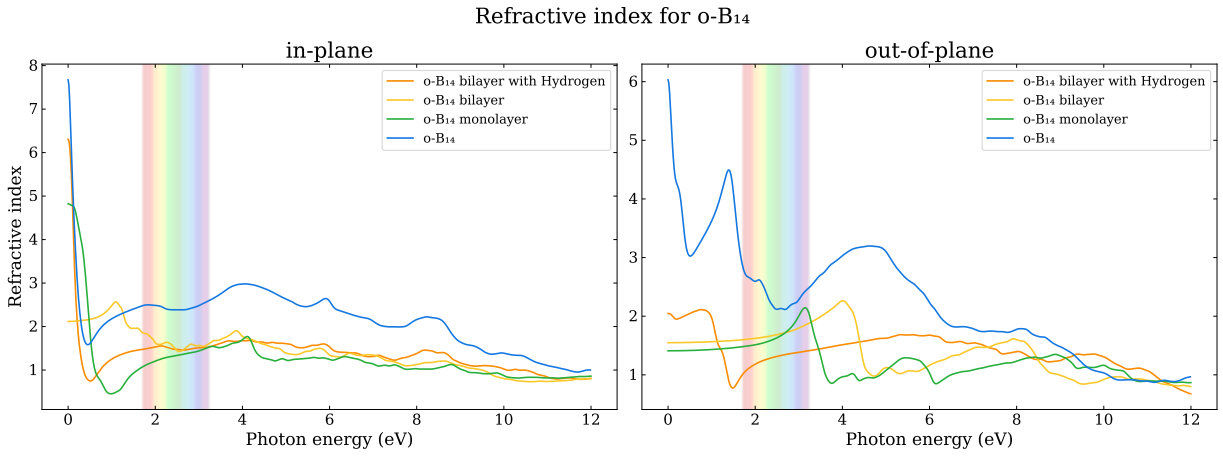

Figure 7: The in-plane and out-of-plane refractive index.

The calculation result of the extinction coefficient:

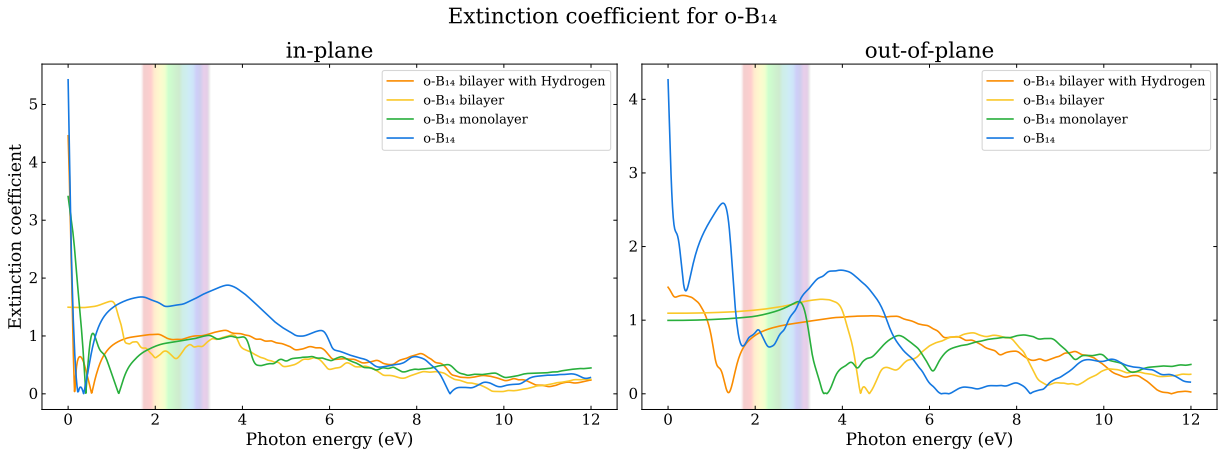

Figure 8: The in-plane and out-of-plane extinction coefficient.

Optical property: reflectivity

Reflectivity describes the reflection of light on the surface of the system. It can be obtained through the Fresnel equations, depending on the material's refractive index and extinction coefficient. For non-magnetic materials, the reflectivity can be expressed using the complex refractive index.

$$N(\omega) = n(\omega) + ik(\omega) \tag{10}$$

According to the Fresnel equations, light is incident perpendicularly to the interface. The reflection coefficient can be expressed as:

$$r = \frac{n - 1 + ik}{n + 1 + ik} \tag{11}$$

The reflectivity R is the squared magnitude of the reflection coefficient r:

$$R = |r|^2 \tag{12}$$

Then, we can get:

$$R(\omega) = \frac{(n(\omega) - 1)^2 + k(\omega)^2}{(n(\omega) + 1)^2 + k(\omega)^2} \tag{13}$$

The part $(n-1)^2 + k^2$ represents the squared amplitude of the reflected wave, taking into account the difference in refractive indices between the incident medium and the material, as well as the material's extinction coefficient. The part $(n+1)^2 + k^2$ represents the total squared amplitude of the incident wave, ensuring that the reflectivity is between 0 and 1.

The calculation result:

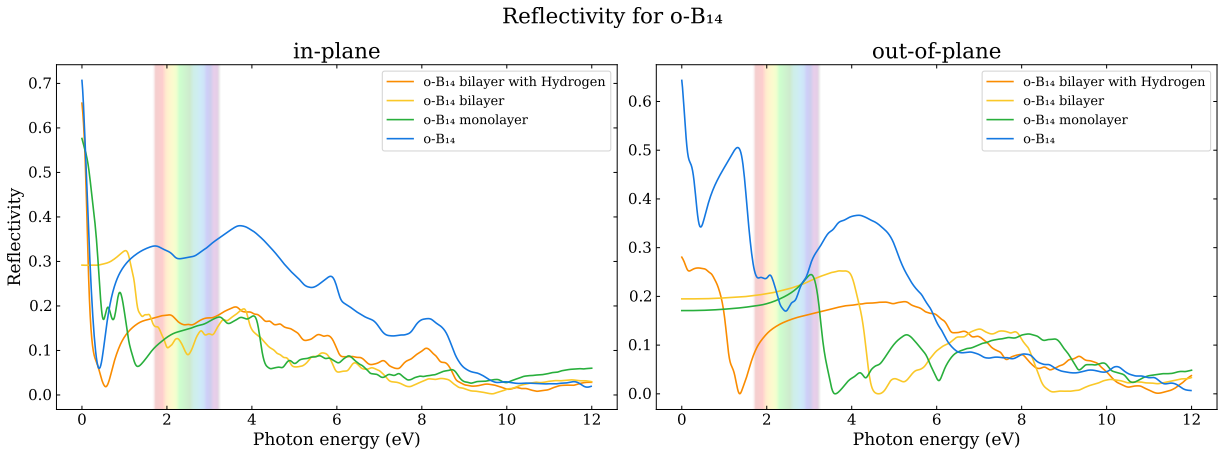

Figure 9: The in-plane and out-of-plane reflectivity.

Optical property: energy loss spectrum

The energy loss function $L(\omega)$ describes the energy loss of electrons as they propagate through the system, specifically reflecting the intensity of energy loss at a given frequency.

$$L(\omega) = \frac{\varepsilon_2(\omega)}{\varepsilon_1^2(\omega) + \varepsilon_2^2(\omega)} \tag{14}$$

This formula directly reflects the energy loss through the imaginary part of the dielectric function $\varepsilon_2(\omega)$, as the imaginary part is directly related to the energy absorbed by the system.

The denominator $\varepsilon_1^2(\omega) + \varepsilon_2^2(\omega)$ ensures the normalization of the function, accounting for the total response of the material to the electric field, thus providing a relative intensity of the energy loss.

The real and imaginary parts of the dielectric function are interrelated through the Kramers-Kronig relations, so the energy loss function depends not only on the imaginary part $\varepsilon_2(\omega)$ but also indirectly on the real part $\varepsilon_1(\omega)$.

The calculation result:

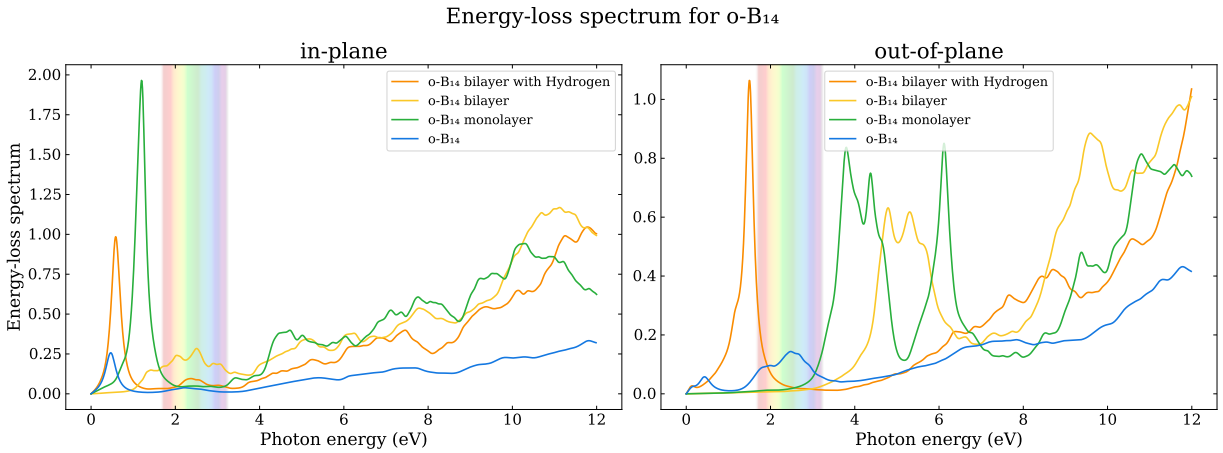

Figure 10: The in-plane and out-of-plane energy loss spectrum.

Thank you